

sult in nanorobots, which will travel in our bloodstream and cure ailments or make observations of a patient's health condition. These robots could also go into other machines to fix them.

At present, however, nanotechnology is more of an upgrade attempt in the field of MEMS, aiming to make components smaller than the micro scale.

Software libraries. "Except for sensors with logical output, signal conditioning and processing need to be performed prior to being used within a behaviour program of the robot. Hence a sensor cannot be thought of as a standalone hardware to produce a meaningful output," comments Jayakrishnan. He cites software libraries as one of the key components that go hand-in-hand with sensors.

"There is a need to provide a consistent software interface with each new generation of hardware. For example, in the majority of imaging applications, the first step is to locate the object of interest. For this, blob analysis is one basic object-locator tool. Once the object has been located, a quality assessment is done, like analysis of colour, texture or geometry of the part. Software libraries such as MIL from Matrox Imaging, Common Vision Blox from Stemmer Imaging, Halcon from MVTec Software and Sapera from Dalsa provide engineers with standard tools for carrying out these sorts of tasks."

Microsoft Kinect. Azad opines that using technologies like Microsoft Kinect also helps to imbibe mobile sensing powers more easily into robots. Kinect is a technology developed by Microsoft, primarily to offer a controller-free gaming and entertainment experience for its Xbox 360 users. Users just need to gesture or speak out their commands. Kinect uses a camera, depth sensor, multi-array microphone and software to interpret the 3D scenes captured.

"Low-cost infrared technology such as that used in Microsoft Kinect's depth sensor can be used to replace



Controller-free gaming using Microsoft Kinect

other expensive single-function sensors, and provides a new route for robotics," says Azad.

Laser-based sensors. Newer laser technologies have contributed a lot to improving the sensory capabilities of robots, especially autonomous ones. Shekar notes, "Previously, sensors like ultrasonic and infrared were used for the study of the environment and obstacle avoidance, but with improvements in the control of laser light and light detection and ranging devices (LIDARs), much more accurate information is obtained, which results in more accurate and efficient operation of robots." Laser-based optical sensing is emerging as a very promising technique.

LIDAR-equipped robots often construct maps of their environment in an *ad hoc* fashion and use it to avoid obstacles. A LIDAR-equipped robotic Boeing AH-6 performed a fully autonomous flight in June 2010. The LIDAR was a basic component of its obstacle-avoidance technique. Around the same time, Dr Srinivasa's team took part in the *Intelligent Ground Vehicle Competition (IGVC)* at Oakland University, Michigan, USA. The objective of the competition was to build a robot that would be able to traverse a path with obstacles in it. One of the common techniques to achieve this is simultaneous localisation and mapping. The team was able to build the robot from scratch, within a time span of three months.

"Getting the funding to buy expensive components like the LIDAR, GPS and the camera was one of the biggest obstacles we had to face. In the

end, we built a system which ran to the capabilities we had expected and we ranked 19th out of 58 teams from four countries in the Design Group A competition," recalls Shekar.

Bio-mimetic systems. If we want robots to be like humans or other natural life forms, why not learn from the nature itself?

"Nanometric sensors have emerged as the result of bio-mimetic systems research. Nature has the best collection of time-tested detection methods, so why not copy from it? The result is bio-mimetic systems that artificially mimic biological processes, using a blend of biology, chemistry, nanotech and engineering. A simple example is the electronic nose that can detect cancer by simply smelling a blood sample. Its natural counterpart is the dog, which is capable of identifying cancer cells in a patient. The process is quite simple to fabricate and is being developed and implemented through elaborate research," says Dr Venkatesh.

Vineet Sahu, another of Dr Srinivasa's students, gives an example to show that such study of nature has inspired the communication skills of robots too: "Swarm robots are one of the developments of such efforts. Swarm robots were an inspiration of industrious social insects like ants and bees.